

ADVANCED JAVA LABSIDDHARTHA ACADEMY OF HIGHER EDUCATION (DEEMED TO BE UNIVERSITY)
(23AI5351) • DEPARTMENT OF AI • BATCH 2025–2029

Week 04: Inheritance, Method Overriding & 'super' Keyword

Student: Trivikram Roll No: 25EU02067 • Lab Record Verified

1. OBJECTIVE / AIM

To demonstrate Single and Multilevel Inheritance, method overriding, dynamic method dispatch, and runtime polymorphism.

2. CONCEPT & THEORY

Inheritance enables a new class to inherit fields and methods from an existing class. Dynamic method dispatch is the mechanism by which a call to an overridden method is resolved at runtime rather than compile time.

3. JAVA SOURCE IMPLEMENTATION

```
// Advanced Java Lab - Experiment 04
// Student: Trivikram | Roll No: 25EU02067
class Employee {
    String name;
    double baseSalary;

    Employee(String name, double baseSalary) {
        this.name = name;
        this.baseSalary = baseSalary;
    }

    void displayRole() {
        System.out.println(name + " | General Role | Salary: Rs." + baseSalary);
    }
}

class SoftwareEngineer extends Employee {
    String techStack;

    SoftwareEngineer(String name, double baseSalary, String techStack) {
        super(name, baseSalary);
        this.techStack = techStack;
    }
}
```

```
@Override
void displayRole() {
    super.displayRole();
    System.out.println("Specialization: " + techStack);
}

}

public class Week4Demo {
    public static void main(String[] args) {
        Employee emp = new SoftwareEngineer("Trivikram", 80000.0, "Java Spring Boot & Microservices");
        emp.displayRole();
    }
}
```

4. COMPILATION & EXECUTION OUTPUT

```
Trivikram | General Role | Salary: Rs.80000.0
Specialization: Java Spring Boot & Microservices
```

